

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question
For the linear function f , the graph of $y = f(x)$ in the xy -plane has a slope of **39** and passes through the point $(0, 0)$. Which equation defines f ?

- Answer**
- A. $f(x) = -39x$
 - B. $f(x) = \frac{1}{39}x$
 - C. $f(x) = x - 39$
 - D. $f(x) = 39x$

Correct Answer: D

Rationale
Choice D is correct. An equation defining a linear function can be written in the form $f(x) = mx + b$, where m is the slope and $(0, b)$ is the y -intercept of the graph of $y = f(x)$ in the xy -plane. It's given that the graph of $y = f(x)$ has a slope of **39**, so $m = 39$. It's also given that the graph of $y = f(x)$ passes through the point $(0, 0)$, so $b = 0$. Substituting **39** for m and **0** for b in $f(x) = mx + b$ yields $f(x) = 39x + 0$, or $f(x) = 39x$. Thus, the equation that defines f is $f(x) = 39x$.

Choice A is incorrect. This equation defines a function whose graph has a slope of **-39**, not **39**.

Choice B is incorrect. This equation defines a function whose graph has a slope of $\frac{1}{39}$, not **39**.

Choice C is incorrect. This equation defines a function whose graph has a slope of **1**, not **39**, and passes through the point $(0, -39)$, not $(0, 0)$.